

Data Science Project Proposal

Outcome 1: Team Collaboration & Project Planning

1. Project Title

“Do low-rated chess players win more games when using common or uncommon openings?”

2. Team Information & Roles

Project Team:

Each team member should have a defined role and responsibilities.

Name	Role(s)	Responsibilities
Anna Bekesi	Analyst, documentation coordinator	Creating python scripts that interrogates our chess data and provides insights that help discover whether or not we can confidently reject our null hypothesis
James Murphy	Team leader, Quality Control	Overseeing the entirety of the project, contributing to all tasks as needed. Checking whether the project is on course and makes sense
Callum Rennie	Data Visualization	Creating visualisations and dashboards that allow us to gain insights into our data and present our findings to an audience

Collaboration Tools:

- **Communication:** Microsoft Teams
 - **Version Control & Documentation:** Git, GitHub, OneDrive.
 - **Project Management:** Microsoft Planner.
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3. Problem Statement & Project Scope

Problem Definition:

Chess Grandmaster Ben Finegold has strongly contended that studying chess openings and the choice of opening used in a game is completely irrelevant for low-rated players (under 2000 Elo). Instead, he states that the moves and tactics used later in the game are what determine match outcomes at this level.

To examine this claim empirically, this project investigates how playing uncommon openings affect match outcome compared to playing common openings.

If a significant difference in win rates is found between the two groups, it would indicate that opening choice does impact the match outcome at lower levels. This would then suggest that picking uncommon openings and studying opening moves gives players an advantage over those who don't, contradicting Finegold's assertion.

Conversely, if no significant difference is found, this would suggest that opening choice does not play a measurable role in performance at lower levels, aligning with Finegold's assertion. However, failing to reject the null hypothesis does not prove that opening choice is irrelevant—it only indicates that our data does not provide strong evidence of an effect.

Statistical Analysis Plan:

To determine whether opening choice impacts match performance, we will conduct two statistical tests on our data to compare the match outcomes of chess games played by players who use uncommon openings (openings used in less than 20% of games) and those who use common openings. These tests will allow us to assess whether there is a statistically significant difference in win rates between the two groups.

Null Hypothesis (H_0): There is no statistically significant difference in the number of wins of low-rated chess players who use common openings and those who use uncommon openings.

Alternative Hypothesis (H_1): A statistically significant difference exists in the win rates of players who use common and those who use uncommon chess openings.

Significance Level (α): 0.05

Test 1

A binomial test will be performed to compare the outcomes of matches where the players are using common openings vs. the players who use uncommon openings. By looking at proportion of successes in our data (ie. a game has been won by using

an uncommon opening), we can determine if they differ from a theoretically expected distribution (playing games normally).

NB exclusion of draws: Since binomial tests require binary outcomes, draws will be excluded from the dataset.

Test 2

Chi-square test of independence will be used to see if there is a relationship between using an uncommon opening and winning the game. By looking at what type of opening a player used, we can then see if they have a bigger chance of winning if using a specific opening according to our dataset.

Additional Considerations: We will check assumptions such as sample size adequacy and potential confounders (e.g., rating differences). We may also perform additional statistical analyses (like logistic regression) if the time permits.

Significance:

This study is important for both the chess community and the broader field of data-driven decision-making:

Challenging Expert Opinions—Grandmaster Ben Finegold's assertion that opening study is irrelevant for low-rated players is widely discussed but lacks empirical testing. By providing data-driven insights, this research adds clarity to an ongoing debate in the chess community.

Practical Guidance for Low-Rated Players—Many beginners and club players struggle with how much time to spend on opening study. If results show a performance difference between common and uncommon openings, it will help players make more informed training decisions.

Implications for Chess Coaching & Training—Coaches often advise students to stick to common openings, but if uncommon openings provide an advantage, this could shift training recommendations.

A Case Study in Data Science & Statistical Analysis—Beyond chess, this project applies real-world data science methods, making it a valuable case study in predictive modelling and hypothesis testing. The methodology could be applied to other domains where expert opinions conflict with empirical evidence.

Stakeholders:

Chess learners and enthusiasts who want to improve their gameplay.

Coaches and trainers looking for data-driven insights on opening strategies.

The broader chess community, including online platforms such as Lichess and Chess.com.

Data science students and teachers who want template projects that can act as a guide to best practice.

Scope:

The project follows the PPDAC cycle, starting with completing the project proposal and brief. The deliverables and timelines are outlined below. The project requirements are completed by the team members.

The following decisions have been made regarding the data and analysis:

- Included in the analysis:

Analysis of Lichess game data, categorization of openings, statistical comparison of performance metrics.

Excluded from the analysis:

Psychological factors influencing chess improvement, individual playing styles, long-term improvement tracking.

Key Questions:

- Does the choice between uncommon and common openings impact the performance of low-rated players?
- How do we assign “responsibility” for an opening to a player
- How significant is the difference in performance between players who play uncommon vs. common openings?
- What factors (e.g., time control, rating differences) influence the results?
- How do we ensure a fair comparison between groups?

4. Objectives & Key Milestones

Project Objectives

- Collaborate effectively within a data science team.
- Develop a structured project plan aligned with the **PPDAC Data Science Lifecycle**.
- Ensure ethical and legal compliance in data handling.
- Set clear deliverables and success criteria.

Key Milestones & Timeline

Milestone	Task Description	Start Date	End Date	Responsible Member(s)
1	Project proposal completed	19/02/2025	24/03/2025	Anna/James/Callum
2	Data processed and ready for analysis	24/03/2025	04/04/2025	Anna/James/Callum
3	Complete EDA and statistical analysis	04/04/2025	21/04/2025	Anna
4	Create Visualisation	21/04/2025	05/05/2025	Callum
5	Submit Report	05/05/2025	19/05/2025	James
6	Presentation	19/05/2025	09/06/2025	Anna/James/Callum

5. Ethical Considerations & Compliance

- **Data Bias:** We will ensure fairness by analyzing data across various rating ranges and avoiding misleading conclusions.
- **Privacy & Security:** Adhere to GDPR and Lichess data usage policies, ensuring no personally identifiable information is used.
- **Transparency:** Document all decisions regarding data usage, methodology, and model selection for reproducibility.

6. Deliverables & Submission Timeline

Deliverable	Description	Due Date
Project Proposal	Define business problem, dataset, and methodology	24/03/2025
Project Plan	Detailed team roles, milestones, and collaboration strategy	24/03/2025